

ChE 311 – Transport Phenomena I –Fall 2024

Rubric Assessment Summary

The assessment of ABET Student Outcome 1 was conducted using the final exam as the primary direct assessment instrument. The exam was scored out of 100 points.

ABET Outcome 1

An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Performance Indicator (PI)	Level 3: Satisfactory >60	Level 2: Developing 45 to 60	Level 1: Unsatisfactory <45
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
6. The uses of appropriate resources needed to solve problems	Uses appropriate resources to locate information needed to solve problems	Uses limited resources to solve problems	Uses no resources to solve problems

FINAL EXAM TOTAL POINTS 100

Average Score	61.50
Standard dev.	18.86

RESULTS BY PERFORMANCE INDICATOR

Mapped SO	1
Average	2.35
Std Dev	0.81
S	11
D	5
U	4

Summary of Findings

The overall student performance on the final exam shows **moderate attainment with noticeable variability**. The **average exam score was 61.50**, with a **standard deviation of 18.86**, indicating a widespread in student achievement levels. This suggests that while some students demonstrated adequate mastery of course material, others struggled with core concepts.

For the mapped performance indicator, the **average rubric score was 2.35 (on a 3-point scale)** with a **standard deviation of 0.81**. This value exceeds the typical attainment benchmark of 2.0, indicating that **SO1 is satisfactorily attained overall**.

However, the distribution of student performance highlights areas of concern:

- **Satisfactory (S): 11 students**
- **Developing (D): 5 students**
- **Unsatisfactory (U): 4 students**

Interpretation

Although the majority of students (16 out of 20, or 80%) achieved at least a developing level, **20% of students did not meet expectations**, indicating persistent learning gaps in some areas of the course.

The relatively high standard deviation in both exam scores and rubric results suggests **uneven student preparedness and inconsistent mastery of key engineering problem-solving skills**.

Overall, students are achieving **acceptable attainment of SO1**, demonstrating the ability to apply engineering principles to solve problems at a satisfactory level. However, the variability in performance and the presence of unsatisfactory outcomes indicate that

some students are not fully developing the depth of understanding required for consistent success.

These results suggest that while instructional methods are effective for many students, they may not adequately support all learners, particularly those who struggle with multi-step problem-solving and application of theoretical concepts.

Recommended Improvements

To enhance student learning and improve consistency of outcomes, the following actions are recommended:

A. Strengthen Foundational Understanding

- Reinforce prerequisite concepts and core principles through brief review modules or refresher materials.
- Emphasize conceptual understanding before introducing complex problem-solving tasks.

B. Increase Formative Assessment Opportunities

- Implement more frequent low-stakes assessments (quizzes, in-class exercises, homework).
- Use formative feedback to identify struggling students earlier in the semester.

C. Enhance Problem-Solving Instruction

- Provide structured, step-by-step solution frameworks for complex engineering problems.
- Incorporate more guided examples and progressively challenging exercises.

D. Improve Student Support Systems

- Offer targeted review sessions and tutoring support for students identified as at risk.
- Encourage greater use of office hours and peer-assisted learning opportunities.

E. Analyze Assessment Alignment

- Conduct a detailed review of exam questions to ensure alignment with learning objectives and instructional emphasis.

Conclusion

While **Student Outcome 1 is achieved, the assessment indicates moderate performance, with significant variability among students.** Continued improvement efforts should focus on strengthening foundational knowledge, increasing formative feedback, and improving instructional support for lower-performing students to ensure more consistent attainment in future course offerings.

FINAL EXAM
TRANSPORT PHENOMENA I, ChE 311, Fall 2024

Closed Book, Time: 90 minutes

1. A spherical pellet of radius $R = 1 \text{ cm}$ and mass of $m = 50 \text{ g}$ is released in a large body of stationary liquid of density $1000 \frac{\text{kg}}{\text{m}^3}$. Estimate the viscosity of the liquid if the terminal velocity of the pellet is $2 \frac{\text{cm}}{\text{s}}$.

$$\text{Friction factor: } f = \left(\sqrt{\frac{24}{Re}} + 0.5407 \right)^2$$

Reynolds number: $Re = \frac{2\rho v_{\infty} R}{\mu}$, where v_{∞} is the terminal velocity

Kinetic force: $F_k = AKf$, where $A = \pi R^2$ and $K = \frac{1}{2} \rho v_{\infty}^2$

$$g = 9.8 \frac{\text{N}}{\text{kg}}$$

Volume of a sphere of radius R is $\frac{4}{3} \pi R^3$.

(40 points)

At terminal velocity: $F_k + \rho g \frac{4}{3} \pi R^3 = mg \rightarrow F_k = 0.45 \text{ N}$

$$F_k = AKf \rightarrow f = \frac{F_k}{\frac{1}{2} \rho v_{\infty}^2 \pi R^2} = 7145$$

$$f = \left(\sqrt{\frac{24}{Re}} + 0.5407 \right)^2 \rightarrow Re = \frac{2\rho v_{\infty} R}{\mu} = 0.0034 \rightarrow \mu = 118 \text{ Pa}\cdot\text{s}$$

2. A liquid with constant density ρ and viscosity μ is situated in the annular space between two long vertical coaxial cylinders of radii R and κR and the liquid is open to the atmosphere at the top. The inner cylinder rotates at a constant angular velocity Ω , while the outer cylinder is fixed.

- 2.1. Using the θ -component of the equation of motion in cylindrical coordinates, find the steady-state velocity profile of the liquid (v_θ as a function of r).

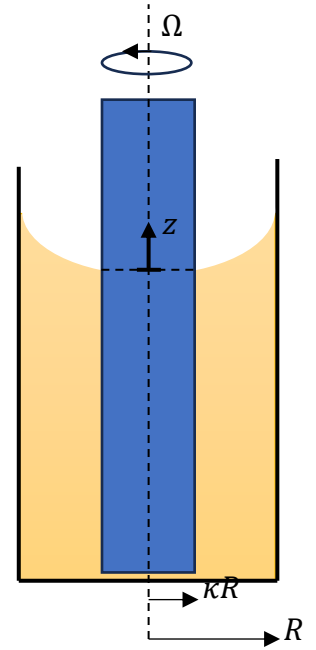
(30 points)

$$\frac{d}{dr} \left[\frac{1}{r} \frac{d}{dr} (r v_\theta) \right] = 0$$

$$v_\theta = C_1 r + \frac{C_2}{r}$$

$$v_\theta(R) = 0 \text{ and } v_\theta(\kappa R) = \kappa R \Omega \rightarrow C_1 = -\frac{\Omega \kappa^2}{1 - \kappa^2} \text{ and } C_2 = \frac{\Omega \kappa^2 R^2}{1 - \kappa^2}$$

$$v_\theta = \frac{\Omega \kappa^2}{1 - \kappa^2} \left(\frac{R^2}{r} - r \right)$$



- 2.2. Using the r -component and the z -component of the equation of motion in cylindrical coordinates, find the shape of the free surface of the liquid (z_s as a function of r).

(30 points)

$$z\text{-component: } 0 = -\frac{\partial p}{\partial z} - \rho g \rightarrow p(r, z = 0) = p_{atm} + \rho g z_s(r) \rightarrow \rho g z_s(r) = p(r, z = 0) - p_{atm}$$

$$r\text{-component: } -\rho \frac{v_\theta^2}{r} = -\frac{\partial p}{\partial r} \rightarrow \rho \frac{v_\theta^2}{r} = \frac{\partial p}{\partial r} = \rho g \frac{dz_s}{dr} \rightarrow \frac{dz_s}{dr} = \frac{v_\theta^2}{g r}$$

$$z_s(r) = \int_{\kappa R}^r dr' \frac{\left[\frac{\Omega \kappa^2}{1 - \kappa^2} \left(\frac{R^2}{r'} - r' \right) \right]^2}{g r'} = \frac{1}{g} \left(\frac{\Omega \kappa^2}{1 - \kappa^2} \right)^2 \int_{\kappa R}^r dr' \frac{\left(\frac{R^2}{r'} - r' \right)^2}{g r'}$$

$$z_s(r) = \frac{1}{2g} \left(\frac{\Omega \kappa^2 R}{1 - \kappa^2} \right)^2 \left[\left(\frac{r}{R} \right)^2 - \left(\frac{r}{R} \right)^{-2} + 4 \log \frac{\kappa R}{r} + \frac{1}{\kappa^2} - \kappa^2 \right]$$